

Munibullah Shah

AI Automation & Backend Engineer

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PROFESSIONAL SUMMARY

AI automation and backend engineer with 5+ years of production software delivery. Operate three LLM agents against live business systems — triaging ~1,000 emails a month over Microsoft Graph and Teams at ~95% classification accuracy, running patient intake for medical practices, and managing Google and Meta ad accounts — and cut one client's scheduled-job model spend by 75% through per-task model routing. Backend work in Node.js/TypeScript and Python across PostgreSQL, Firestore and Google Cloud, including IAM and service-account design. Previously shipped 10+ mobile applications to the App Store and Google Play, including a government emergency-response platform now past 500K downloads.

TECHNICAL SKILLS

AI & Agents: LLM orchestration, autonomous tool calling, multi-step agent loops, agent guardrails, prompt and context engineering, structured output, classification taxonomy design, output evaluation, model routing and inference cost optimisation, GPT-5, GPT-4o mini, Google Gemini, Vercel AI SDK, OpenClaw

Retrieval: Embeddings, vector similarity matching, document ingestion, chunking, grounded generation

Backend: Node.js, TypeScript, Express, Python, REST APIs, JWT and OAuth, Stripe, webhooks, Firebase Cloud Functions

Data: PostgreSQL, MongoDB, Firestore, SQL, pandas, ETL pipelines, star schema modelling, data quality gates

Cloud & Infrastructure: Google Cloud (IAM roles and policy bindings, service accounts, OAuth scopes, domain-wide delegation), Firebase, Docker, Traefik, Dokploy, DigitalOcean, cron scheduling

Integrations: Microsoft Graph, Google Chat, Google Ads API, Meta Ads API, Palmetto LightReach, Maton API, ERPNext, Typebot, PostHog

Mobile: Flutter, Dart, BLOC/Cubit, Provider, Clean Architecture, RevenueCat, Android (Java)

PROFESSIONAL EXPERIENCE

Full Stack Engineer — AI Automation & Mobile

Ambeego, Remote | 09/2025 – Present

- Build and operate AI-driven patient intake automation for a white-labelled clinical platform serving private medical practices; live site now processes 2-3 new patient registrations daily and removes 7-10 minutes of appointment scheduling and initial doctor screening per patient.
- Engineer LLM prompts producing separate clinical and patient-facing summaries, written back as records linked to the correct patient in an ERPNext Healthcare CRM.
- Deploy and debug self-hosted infrastructure on Dokploy, Docker and Traefik, including a WhatsApp intake channel on a dedicated DigitalOcean instance; resolved router-name collisions and inter-service DNS conflicts.
- Automate CRM role and permission configuration by script across multi-tenant infrastructure.
- Delivered the Baby Resume application end to end — Flutter (iOS and Android), Node.js backend and Firebase infrastructure, with Clean Architecture and BLOC/Cubit across 37+ screens.

Freelance AI Automation & Backend Engineer

Upwork and Direct Clients, Remote | 01/2021 – Present

- Operate three AI assistants in production, including an email triage assistant handling 30-40 emails daily (~1,000 per month) over Microsoft Graph and Teams for a Belgian managed-services provider, and a Google/Meta ads management agent via the Maton API gateway.
- Defined the classification taxonomy the triage agent is graded against - quotation requests to business, colleague queries to team question, and further categories - and review agent output against that rubric; classification currently runs ~95% correct.
- Reduced a client's scheduled-job LLM spend 75% - from \$50 to \$10-14 per month - by profiling per-task requirements and routing work from GPT-5 down to GPT-4o mini; adaptive per-task selection in progress.
- Built a Google Workspace operations assistant for a commercial HVAC contractor: email summaries delivered through a deployed Google Chat app, with Google Ads, Meta Ads and Palmetto LightReach project status unified into one tool surface.
- Designed the cloud identity layer behind it — GCP service accounts, OAuth scope selection, IAM roles and policy bindings, and domain-wide delegation for Workspace-wide Chat app authorisation.
- Hardened agents after live production incidents: confidentiality rules enforced in the agent system layer, tool-layer refusal guards, and credential rotation and separation across agents.

- Delivered an autonomous tool-calling Firebase Cloud Function using the Vercel AI SDK and Google Gemini, with embedding-based content matching and schema-validated JSON output.
- Built the Healix AI application (Flutter, Python Cloud Functions, OpenAI GPT) with 17+ HTTPS endpoints, offline chat caching and scheduled push notifications; 10K+ downloads on the App Store.
- Maintained a 5.0 rating across all Upwork contracts with multiple repeat clients.

Flutter Developer (Contract)

SuperHero Inc., Islamabad | 02/2024 – 07/2025

- Led development of the SuperHero eSIM application end to end, shipping to the App Store and Google Play with coverage across 150+ countries.
- Integrated Stripe end to end — checkout, payment flows and webhook handling — for secure cross-border transactions.
- Built and maintained REST integrations for eSIM provisioning and data plan lifecycle management.

Software Engineer — Android and Flutter

Hoboetech, Lahore | 09/2022 – 12/2023

- Shipped 5+ mobile applications from requirements to store release, including the Rescue emergency response app for the Department of Emergency Services, Government of Punjab, now past 500K downloads.
- Architected Firebase integrations across Auth, Realtime Database, Firestore and Cloud Functions.
- Built mapping features including polylines, geofencing and real-time tracking using Google Maps and Mapbox.
- Owned 3 applications end to end as sole developer, from architecture through release.

Flutter Developer Intern

IOI Technologies, Lahore | 02/2021 – 09/2021

- Delivered production UI screens and applied GetX and Provider state management in a live codebase.

SELECTED PROJECTS

Production AI Agent Suite. Two LLM agents on live business systems: email triage over Microsoft Graph surfaced in Teams (~1,000 emails a month, ~95% correctly classified against a defined category rubric), and a Google/Meta ads agent through an API gateway. Model routing cut scheduled-job spend 75%; guardrails and least-privilege tool surface designed after real production incidents.

AI Operations Assistant, Commercial HVAC. Google Workspace assistant summarising inbound email into a deployed Google Chat app, with Google Ads, Meta Ads and Palmetto LightReach project status integrations. Full GCP service account, OAuth scope and IAM policy design.

Clinical Intake Automation Platform. Multi-practice conversational intake with LLM-generated clinical and patient-facing summaries written into an ERPNext Healthcare CRM, on self-hosted Docker, Traefik and Dokploy infrastructure. Live at 2-3 patient registrations daily, saving 7-10 minutes of clinician time each.

Autonomous Tool-Calling Clinic Assistant. Firebase Cloud Function running a multi-step Gemini agent loop via the Vercel AI SDK, with embedding-based content matching and schema-validated structured output.

Toast POS ETL Pipeline. Python pipeline ingesting POS sales data into PostgreSQL through a four-phase flow (raw JSONB, staging, star schema warehouse, analytics views) with SCD dimension tracking, data quality gates and daily cron scheduling.

SuperHero eSIM. Travel eSIM application for 150+ countries with Stripe payments and provisioning API integration. 5.0 on the App Store, 10K+ downloads on Google Play.

Rescue Emergency Services Suite. Three Flutter apps for the Department of Emergency Services, Government of Punjab, covering citizen emergency response, field staff attendance and rescue worker operations. 500K+ downloads.

EDUCATION

Bachelor of Science in Software Engineering

COMSATS University Islamabad, Lahore Campus | 2021

CERTIFICATIONS

Google Cloud Skills Boost — six badges including Google Cloud Fundamentals: Core Infrastructure, Implementing Cloud Load Balancing for Compute Engine, Exploring Data Transformation with Google Cloud, and Digital Transformation with Google Cloud (07/2020 – 05/2025).